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2026-03-19

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1.1

- SMTP-

1.1

- SMTP-
- Postfix

2

2.

- server

```
[user@server.user.net server]$ sudo -i
[sudo] password for user:
[root@server.user.net ~]# dnf -y install postfix
Last metadata expiration check: 2:30:23 ago on Sat 31 Jan 2026 07:14:21 PM UTC.
Package postfix-2:3.5.25-1.el9.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@server.user.net ~]# dnf -y install s-nail
Last metadata expiration check: 2:30:56 ago on Sat 31 Jan 2026 07:14:21 PM UTC.
Package s-nail-14.9.22-6.el9.x86_64 is already installed.
Dependencies resolved.
================================================================================================================================
 Package                    Architecture                Version                          Repository                        Size
================================================================================================================================
Upgrading:
 s-nail                     x86_64                      14.9.22-9.el9_7                  appstream                        619 k

Transaction Summary
================================================================================================================================
Upgrade  1 Package

Total download size: 619 k
Downloading Packages:
s-nail-14.9.22-9.el9_7.x86_64.rpm                                                   1.1 MB/s | 619 kB     00:00
--------------------------------------------------------------------------------------------------------------------------------
Total                                                                               608 kB/s | 619 kB     00:01
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                                                                     1/1
  Running scriptlet: s-nail-14.9.22-9.el9_7.x86_64                                                                       1/2
  Upgrading        : s-nail-14.9.22-9.el9_7.x86_64                                                                       1/2
  Running scriptlet: s-nail-14.9.22-9.el9_7.x86_64                                                                       1/2
```

2.1

- server
- sudo -i

```
[user@server.user.net server]$ sudo -i
[sudo] password for user:
[root@server.user.net ~]# dnf -y install postfix
Last metadata expiration check: 2:30:23 ago on Sat 31 Jan 2026 07:14:21 PM UTC.
Package postfix-2:3.5.25-1.el9.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@server.user.net ~]# dnf -y install s-nail
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Package s-nail-14.9.22-6.el9.x86_64 is already installed.
Dependencies resolved.
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Upgrading:
 s-nail                     x86_64                      14.9.22-9.el9_7                  appstream                        619 k

Transaction Summary
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Upgrade  1 Package

Total download size: 619 k
Downloading Packages:
s-nail-14.9.22-9.el9_7.x86_64.rpm                                                   1.1 MB/s | 619 kB     00:00
--------------------------------------------------------------------------------------------------------------------------------
Total                                                                               608 kB/s | 619 kB     00:01
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                                                                     1/1
  Running scriptlet: s-nail-14.9.22-9.el9_7.x86_64                                                                       1/2
  Upgrading        : s-nail-14.9.22-9.el9_7.x86_64                                                                       1/2
  Running scriptlet: s-nail-14.9.22-9.el9_7.x86_64                                                                       1/2
```

2.1

- server
- sudo -i
- root@server.bahis.net

```
[user@server.user.net server]$ sudo -i
[sudo] password for user:
[root@server.user.net ~]# dnf -y install postfix
Last metadata expiration check: 2:30:23 ago on Sat 31 Jan 2026 07:14:21 PM UTC.
Package postfix-2:3.5.25-1.el9.x86_64 is already installed.
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Last metadata expiration check: 2:30:56 ago on Sat 31 Jan 2026 07:14:21 PM UTC.
Package s-nail-14.9.22-6.el9.x86_64 is already installed.
Dependencies resolved.
================================================================================================================================
 Package                    Architecture            Version                         Repository                          Size
================================================================================================================================
Upgrading:
 s-nail                     x86_64                  14.9.22-9.el9_7                 appstream                          619 k

Transaction Summary
================================================================================================================================
Upgrade  1 Package

Total download size: 619 k
Downloading Packages:
s-nail-14.9.22-9.el9_7.x86_64.rpm                                                   1.1 MB/s | 619 kB     00:00
--------------------------------------------------------------------------------------------------------------------------------
Total                                                                               608 kB/s | 619 kB     00:01
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                                                                     1/1
  Running scriptlet: s-nail-14.9.22-9.el9_7.x86_64                                                                       1/2
  Upgrading        : s-nail-14.9.22-9.el9_7.x86_64                                                                       1/2
  Running scriptlet: s-nail-14.9.22-9.el9_7.x86_64                                                                       1/2
```

2.2 Postfix s-nail

- postfix s-nail dnf

```
[root@server.user.net ~]# dnf -y install postfix
Last metadata expiration check: 2:35:13 ago on Sat 31 Jan 2026 07:14:21 PM UTC.
Package postfix-2:3.5.25-1.el9.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@server.user.net ~]# dnf -y install s-nail
Last metadata expiration check: 2:35:40 ago on Sat 31 Jan 2026 07:14:21 PM UTC.
Package s-nail-14.9.22-9.el9_7.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@server.user.net ~]#
```

2: postfix s-nail dnf

2.2 Postfix s-nail

- postfix s-nail dnf
-

```
[root@server.user.net ~]# dnf -y install postfix
Last metadata expiration check: 2:35:13 ago on Sat 31 Jan 2026 07:14:21 PM UTC.
Package postfix-2:3.5.25-1.el9.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@server.user.net ~]# dnf -y install s-nail
Last metadata expiration check: 2:35:40 ago on Sat 31 Jan 2026 07:14:21 PM UTC.
Package s-nail-14.9.22-9.el9_7.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@server.user.net ~]#
```

2: postfix s-nail dnf

2.2 Postfix s-nail

- postfix s-nail dnf
-
-

```
[root@server.user.net ~]# dnf -y install postfix
Last metadata expiration check: 2:35:13 ago on Sat 31 Jan 2026 07:14:21 PM UTC.
Package postfix-2:3.5.25-1.el9.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@server.user.net ~]# dnf -y install s-nail
Last metadata expiration check: 2:35:40 ago on Sat 31 Jan 2026 07:14:21 PM UTC.
Package s-nail-14.9.22-9.el9_7.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@server.user.net ~]#
```

2: postfix s-nail dnf

SMTP

- smtp firewall-cmd

```
[root@server.user.net ~]# firewall-cmd --add-service=smtp
Warning: ALREADY_ENABLED: 'smtp' already in 'public'
success
[root@server.user.net ~]# firewall-cmd --add-service=smtp --permanent
Warning: ALREADY_ENABLED: smtp
success
[root@server.user.net ~]# firewall-cmd --list-services
cockpit dhcpv6-client ntp samba smtp ssh ssh-custom
[root@server.user.net ~]#
```

3: SMTP firewalld

SMTP

- smtp firewall-cmd
-

```
[root@server.user.net ~]# firewall-cmd --add-service=smtp
Warning: ALREADY_ENABLED: 'smtp' already in 'public'
success
[root@server.user.net ~]# firewall-cmd --add-service=smtp --permanent
Warning: ALREADY_ENABLED: smtp
success
[root@server.user.net ~]# firewall-cmd --list-services
cockpit dhcpv6-client ntp samba smtp ssh ssh-custom
[root@server.user.net ~]#
```

3: SMTP firewalld

SMTP

- smtp firewall-cmd
-
- --list-services smtp

```
[root@server.user.net ~]# firewall-cmd --add-service=smtp
Warning: ALREADY_ENABLED: 'smtp' already in 'public'
success
[root@server.user.net ~]# firewall-cmd --add-service=smtp --permanent
Warning: ALREADY_ENABLED: smtp
success
[root@server.user.net ~]# firewall-cmd --list-services
cockpit dhcpv6-client ntp samba smtp ssh ssh-custom
[root@server.user.net ~]#
```

3: SMTP firewalld

- `restorecon -vR /etc`

```
[root@server.user.net ~]# firewall-cmd --add-service=smtp
Warning: ALREADY_ENABLED: 'smtp' already in 'public'
success
[root@server.user.net ~]# firewall-cmd --add-service=smtp --permanent
Warning: ALREADY_ENABLED: smtp
success
[root@server.user.net ~]# firewall-cmd --list-services
cockpit dhcpv6-client ntp samba smtp ssh ssh-custom
[root@server.user.net ~]#
```

4: SELinux

- `restorecon -vR /etc`
-

```
[root@server.user.net ~]# firewall-cmd --add-service=smtp
Warning: ALREADY_ENABLED: 'smtp' already in 'public'
success
[root@server.user.net ~]# firewall-cmd --add-service=smtp --permanent
Warning: ALREADY_ENABLED: smtp
success
[root@server.user.net ~]# firewall-cmd --list-services
cockpit dhcpv6-client ntp samba smtp ssh ssh-custom
[root@server.user.net ~]#
```

4: SELinux

- `restorecon -vR /etc`
-
-

```
[root@server.user.net ~]# firewall-cmd --add-service=smtp
Warning: ALREADY_ENABLED: 'smtp' already in 'public'
success
[root@server.user.net ~]# firewall-cmd --add-service=smtp --permanent
Warning: ALREADY_ENABLED: smtp
success
[root@server.user.net ~]# firewall-cmd --list-services
cockpit dhcpv6-client ntp samba smtp ssh ssh-custom
[root@server.user.net ~]#
```

4: SELinux

- `systemctl enable postfix`

```
[root@server.user.net ~]# postconf myorigin
myorigin = $mydomain
[root@server.user.net ~]# postconf mydomain
mydomain = user.net
[root@server.user.net ~]# postconf -e 'myorigin = $mydomain'
> postconf myorigin
>
```

5: Postfix

Postfix

- `systemctl enable postfix`
- `systemctl start postfix`

```
[root@server.user.net ~]# postconf myorigin
myorigin = $mydomain
[root@server.user.net ~]# postconf mydomain
mydomain = user.net
[root@server.user.net ~]# postconf -e 'myorigin = $mydomain'
> postconf myorigin
>
```

5: Postfix

Postfix

- `systemctl enable postfix`
- `systemctl start postfix`
- `systemctl status postfix` — `active (running)`

```
[root@server.user.net ~]# postconf myorigin
myorigin = $mydomain
[root@server.user.net ~]# postconf mydomain
mydomain = user.net
[root@server.user.net ~]# postconf -e 'myorigin = $mydomain'
> postconf myorigin
>
```

5: Postfix

- postconf

```
[user@server.user.net server]$ postfix check
postfix: error: to submit mail, use the Postfix sendmail command
postfix: fatal: the postfix command is reserved for the superuser
[user@server.user.net server]$ systemctl reload postfix
[user@server.user.net server]$ postconf -n
alias_database = hash:/etc/aliases
alias_maps = hash:/etc/aliases
command_directory = /usr/sbin
compatibility_level = 2
daemon_directory = /usr/libexec/postfix
data_directory = /var/lib/postfix
debug_peer_level = 2
debugger_command = PATH=/bin:/usr/bin:/usr/local/bin:/usr/X11R6/bin ddd $daemon_directory/$process_name $process_id & sleep 5
html_directory = no
inet_interfaces = localhost
inet_protocols = ipv4
mail_owner = postfix
mailq_path = /usr/bin/mailq.postfix
manpage_directory = /usr/share/man
meta_directory = /etc/postfix
mydestination = $myhostname, localhost.$mydomain, localhost
mydomain = user.net
myorigin = $mydomain
```

6: postconf —

Postfix

- postconf
-

```
[user@server.user.net server]$ postfix check
postfix: error: to submit mail, use the Postfix sendmail command
postfix: fatal: the postfix command is reserved for the superuser
[user@server.user.net server]$ systemctl reload postfix
[user@server.user.net server]$ postconf -n
alias_database = hash:/etc/aliases
alias_maps = hash:/etc/aliases
command_directory = /usr/sbin
compatibility_level = 2
daemon_directory = /usr/libexec/postfix
data_directory = /var/lib/postfix
debug_peer_level = 2
debugger_command = PATH=/bin:/usr/bin:/usr/local/bin:/usr/X11R6/bin ddd $daemon_directory/$process_name $process_id & sleep 5
html_directory = no
inet_interfaces = localhost
inet_protocols = ipv4
mail_owner = postfix
mailq_path = /usr/bin/mailq.postfix
manpage_directory = /usr/share/man
meta_directory = /etc/postfix
mydestination = $myhostname, localhost.$mydomain, localhost
mydomain = user.net
myorigin = $mydomain
```

6: postconf —

Postfix

- postconf
-
-

```
[user@server.user.net server]$ postfix check
postfix: error: to submit mail, use the Postfix sendmail command
postfix: fatal: the postfix command is reserved for the superuser
[user@server.user.net server]$ systemctl reload postfix
[user@server.user.net server]$ postconf -n
alias_database = hash:/etc/aliases
alias_maps = hash:/etc/aliases
command_directory = /usr/sbin
compatibility_level = 2
daemon_directory = /usr/libexec/postfix
data_directory = /var/lib/postfix
debug_peer_level = 2
debugger_command = PATH=/bin:/usr/bin:/usr/local/bin:/usr/X11R6/bin ddd $daemon_directory/$process_name $process_id & sleep 5
html_directory = no
inet_interfaces = localhost
inet_protocols = ipv4
mail_owner = postfix
mailq_path = /usr/bin/mailq.postfix
manpage_directory = /usr/share/man
meta_directory = /etc/postfix
mydestination = $myhostname, localhost.$mydomain, localhost
mydomain = user.net
myorigin = $mydomain
```

6: postconf —

2.7 myorigin mydomain

- `myorigin = $myhostname`

```
[user@server.user.net server]$ postconf -e 'mydomain = user.net'
postconf: fatal: open /etc/postfix/main.cf.tmp: Permission denied
[user@server.user.net server]$ postconf inet_protocols
inet_protocols = ipv4
[user@server.user.net server]$ postconf -e 'inet_protocols = ipv4
>
```

7: myorigin

2.7 myorigin mydomain

- myorigin = $myhostname
- mydomain = bahis.net

```
[user@server.user.net server]$ postconf -e 'mydomain = user.net'
postconf: fatal: open /etc/postfix/main.cf.tmp: Permission denied
[user@server.user.net server]$ postconf inet_protocols
inet_protocols = ipv4
[user@server.user.net server]$ postconf -e 'inet_protocols = ipv4
>
```

7: myorigin

2.7 myorigin mydomain

- myorigin = $myhostname
- mydomain = bahis.net
- myorigin = $mydomain

```
[user@server.user.net server]$ postconf -e 'mydomain = user.net'
postconf: fatal: open /etc/postfix/main.cf.tmp: Permission denied
[user@server.user.net server]$ postconf inet_protocols
inet_protocols = ipv4
[user@server.user.net server]$ postconf -e 'inet_protocols = ipv4
>
```

7: myorigin

2.7 myorigin mydomain

- myorigin = $myhostname
- mydomain = bahis.net
- myorigin = $mydomain
- postfix check systemctl reload postfix

```
[user@server.user.net server]$ postconf -e 'mydomain = user.net'
postconf: fatal: open /etc/postfix/main.cf.tmp: Permission denied
[user@server.user.net server]$ postconf inet_protocols
inet_protocols = ipv4
[user@server.user.net server]$ postconf -e 'inet_protocols = ipv4
>
```

7: myorigin

2.7 myorigin mydomain

- myorigin = $myhostname
- mydomain = bahis.net
- myorigin = $mydomain
- postfix check systemctl reload postfix
- postconf -n

```
[user@server.user.net server]$ postconf -e 'mydomain = user.net'
postconf: fatal: open /etc/postfix/main.cf.tmp: Permission denied
[user@server.user.net server]$ postconf inet_protocols
inet_protocols = ipv4
[user@server.user.net server]$ postconf -e 'inet_protocols = ipv4
>
```

7: myorigin

- `mydomain = bahis.net`

```
[user@server.user.net server]$ echo .| mail -s test1 user@server.user.net
[user@server.user.net server]$
```

8: mydomain inet_protocols

IPv6

- mydomain = bahis.net
- inet_protocols = all

```
[user@server.user.net server]$ echo .| mail -s test1 user@server.user.net
[user@server.user.net server]$
```

8: mydomain inet_protocols

IPv6

- mydomain = bahis.net
- inet_protocols = all
- inet_protocols = ipv4

```
[user@server.user.net server]$ echo .| mail -s test1 user@server.user.net
[user@server.user.net server]$
```

8: mydomain inet_protocols

IPv6

- mydomain = bahis.net
- inet_protocols = all
- inet_protocols = ipv4
- postfix check systemctl reload postfix

```
[user@server.user.net server]$ echo .| mail -s test1 user@server.user.net
[user@server.user.net server]$
```

8: mydomain inet_protocols

2.8.1 2.

- /var/log/maillog

```
[user@server.user.net server]$ sudo -i
[sudo] password for user:
[root@server.user.net ~]# dnf -y install postfix
Last metadata expiration check: 3:00:24 ago on Sat 31 Jan 2026 07:14:21 PM UTC.
Package postfix-2:3.5.25-1.el9.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@server.user.net ~]# dnf -y install s-nail
Last metadata expiration check: 3:00:49 ago on Sat 31 Jan 2026 07:14:21 PM UTC.
Package s-nail-14.9.22-9.el9_7.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@server.user.net ~]#
```

9: maillog

2.8.1 2.

- /var/log/maillog
- status=sent (delivered to mailbox)

```
[user@server.user.net server]$ sudo -i
[sudo] password for user:
[root@server.user.net ~]# dnf -y install postfix
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Package s-nail-14.9.22-9.el9_7.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@server.user.net ~]#
```

9: maillog

2.8.1 2.

- /var/log/maillog
- status=sent (delivered to mailbox)
- removed

```
[user@server.user.net server]$ sudo -i
[sudo] password for user:
[root@server.user.net ~]# dnf -y install postfix
Last metadata expiration check: 3:00:24 ago on Sat 31 Jan 2026 07:14:21 PM UTC.
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Package s-nail-14.9.22-9.el9_7.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@server.user.net ~]#
```

9: maillog

2.8.1 2.

- /var/log/maillog
- status=sent (delivered to mailbox)
- removed
- /var/spool/mail

```
[user@server.user.net server]$ sudo -i
[sudo] password for user:
[root@server.user.net ~]# dnf -y install postfix
Last metadata expiration check: 3:00:24 ago on Sat 31 Jan 2026 07:14:21 PM UTC.
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Dependencies resolved.
Nothing to do.
Complete!
[root@server.user.net ~]#
```

9: maillog

2.8.1 2.

- /var/log/maillog
- status=sent (delivered to mailbox)
- removed
- /var/spool/mail
-

```
[user@server.user.net server]$ sudo -i
[sudo] password for user:
[root@server.user.net ~]# dnf -y install postfix
Last metadata expiration check: 3:00:24 ago on Sat 31 Jan 2026 07:14:21 PM UTC.
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Package s-nail-14.9.22-9.el9_7.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@server.user.net ~]#
```

9: maillog

2.9 Postfix s-nail

- client postfix s-nail

```
[root@server.user.net ~]# echo .| mail -s test2 user@user.net
[root@server.user.net ~]# tail -f /var/log/maillog
Jan 31 22:20:06 server postfix/cleanup[48537]: 969198266F5: message-id=<20260131222006.969198266F5@server.user.net>
Jan 31 22:20:06 server postfix/qmgr[48341]: 969198266F5: from=<root@user.net>, size=305, nrcpt=1 (queue active)
Jan 31 22:20:06 server postfix/smtp[48539]: warning: run-time library vs. compile-time header version mismatch: OpenSSL 3.5.0 may not b
e compatible with OpenSSL 3.2.0
Jan 31 22:20:06 server postfix/smtp[48539]: 969198266F5: to=<user@user.net>, relay=none, delay=0.41, delays=0.14/0.08/0.19/0, dsn=5.4.4
, status=bounced (Host or domain name not found. Name service error for name=user.net type=A: Host not found)
```

10: postfix

```
[user@server.user.net server]$ postconf inet_interfaces
inet_interfaces = localhost
[user@server.user.net server]$ postconf mynetworks
mynetworks = 127.0.0.1/32
[user@server.user.net server]$ postconf -e 'inet_interfaces = all'
postconf: fatal: open /etc/postfix/main.cf.tmp: Permission denied
[user@server.user.net server]$ postfix check
postfix: error: to submit mail, use the Postfix sendmail command
postfix: fatal: the postfix command is reserved for the superuser
[user@server.user.net server]$ systemctl reload postfix
[user@server.user.net server]$ systemctl stop postfix
[user@server.user.net server]$ systemctl start postfix
[user@server.user.net server]$
```

2.9 Postfix s-nail

- client postfix s-nail
-

```
[root@server.user.net ~]# echo .| mail -s test2 user@user.net
[root@server.user.net ~]# tail -f /var/log/maillog
Jan 31 22:20:06 server postfix/cleanup[48537]: 969198266F5: message-id=<20260131222006.969198266F5@server.user.net>
Jan 31 22:20:06 server postfix/qmgr[48341]: 969198266F5: from=<root@user.net>, size=305, nrcpt=1 (queue active)
Jan 31 22:20:06 server postfix/smtp[48539]: warning: run-time library vs. compile-time header version mismatch: OpenSSL 3.5.0 may not b
e compatible with OpenSSL 3.2.0
Jan 31 22:20:06 server postfix/smtp[48539]: 969198266F5: to=<user@user.net>, relay=none, delay=0.41, delays=0.14/0.08/0.19/0, dsn=5.4.4
, status=bounced (Host or domain name not found. Name service error for name=user.net type=A: Host not found)
```

10: postfix

```
[user@server.user.net server]$ postconf inet_interfaces
inet_interfaces = localhost
[user@server.user.net server]$ postconf mynetworks
mynetworks = 127.0.0.1/32
[user@server.user.net server]$ postconf -e 'inet_interfaces = all'
postconf: fatal: open /etc/postfix/main.cf.tmp: Permission denied
[user@server.user.net server]$ postfix check
postfix: error: to submit mail, use the Postfix sendmail command
postfix: fatal: the postfix command is reserved for the superuser
[user@server.user.net server]$ systemctl reload postfix
[user@server.user.net server]$ systemctl stop postfix
[user@server.user.net server]$ systemctl start postfix
[user@server.user.net server]$
```

- inet_protocols

```
[user@server.user.net server]$ echo .| mail -s test2 user@user.net
[user@server.user.net server]$
```

12: inet_protocols Postfix

- inet_protocols
- inet_protocols = ipv4

```
[user@server.user.net server]$ echo .| mail -s test2 user@user.net
[user@server.user.net server]$
```

12: inet_protocols Postfix

- inet_protocols
- inet_protocols = ipv4
-

```
[user@server.user.net server]$ echo .| mail -s test2 user@user.net
[user@server.user.net server]$
```

12: inet_protocols Postfix

Postfix

- inet_protocols
- inet_protocols = ipv4
-
- Postfix

```
[user@server.user.net server]$ echo .| mail -s test2 user@user.net
[user@server.user.net server]$
```

12: inet_protocols Postfix

2.11

- test2 bahis@server.bahis.net

```
[user@server.user.net server]$ postqueue -p
Mail queue is empty
[user@server.user.net server]$
```

13: test2

2.12

- `inet_interfaces = localhost`

```
[user@server.user.net ~]$ cd /vagrant/provision/server
[user@server.user.net server]$ touch mail.sh
[user@server.user.net server]$ chmod +x mail.sh
[user@server.user.net server]$
```

14: inet_interfaces mynetworks

2.12

- `inet_interfaces = localhost`
- `mynetworks = 127.0.0.1/32`

```
[user@server.user.net ~]$ cd /vagrant/provision/server
[user@server.user.net server]$ touch mail.sh
[user@server.user.net server]$ chmod +x mail.sh
[user@server.user.net server]$
```

14: inet_interfaces mynetworks

2.12

- `inet_interfaces = localhost`
- `mynetworks = 127.0.0.1/32`
-

```
[user@server.user.net ~]$ cd /vagrant/provision/server
[user@server.user.net server]$ touch mail.sh
[user@server.user.net server]$ chmod +x mail.sh
[user@server.user.net server]$
```

14: inet_interfaces mynetworks

- `inet_interfaces = all`

```
[user@server.user.net ~]$ cd /vagrant/provision/server
[user@server.user.net server]$ touch mail.sh
[user@server.user.net server]$ chmod +x mail.sh
[user@server.user.net server]$
```

15: Postfix

2.13

- `inet_interfaces = all`
- `mynetworks = 127.0.0.0/8, 192.168.0.0/16`

```
[user@server.user.net ~]$ cd /vagrant/provision/server
[user@server.user.net server]$ touch mail.sh
[user@server.user.net server]$ chmod +x mail.sh
[user@server.user.net server]$
```

15: Postfix

2.13

- `inet_interfaces = all`
- `mynetworks = 127.0.0.0/8, 192.168.0.0/16`
- `postfix check`,

```
[user@server.user.net ~]$ cd /vagrant/provision/server
[user@server.user.net server]$ touch mail.sh
[user@server.user.net server]$ chmod +x mail.sh
[user@server.user.net server]$
```

15: Postfix

2.14

-

```
Feb 11 20:34:16 server postfix/smtpd[10156]: disconnect from client.bahis.net[192.168.1.30]
 ehlo=2 starttls=1 mail=1 rcpt=1 data=1 quit=1 commands=7
Feb 11 20:34:16 server postfix/local[10161]: 794CE14A9: to=<bahis@server.bahis.net>, relay=
local, delay=0.02, delays=0.01/0.01/0/0, dsn=2.0.0, status=sent (delivered to mailbox)
Feb 11 20:34:16 server postfix/qmgr[10155]: 794CE14A9: removed
^C
[root@server.bahis.net ~]# tail -f /var/log/maillog
Feb 11 20:33:43 server postfix/postfix-script[10151]: starting the Postfix mail system
Feb 11 20:33:43 server postfix/master[10153]: daemon started -- version 3.5.25, configurati
on /etc/postfix
Feb 11 20:34:16 server postfix/smtpd[10156]: warning: run-time library vs. compile-time hea
der version mismatch: OpenSSL 3.5.0 may not be compatible with OpenSSL 3.2.0
```

16:

2.14

-
- : Received: from client.bahis.net (192.168.1.30)

```
Feb 11 20:34:16 server postfix/smtpd[10156]: disconnect from client.bahis.net[192.168.1.30]
 ehlo=2 starttls=1 mail=1 rcpt=1 data=1 quit=1 commands=7
Feb 11 20:34:16 server postfix/local[10161]: 794CE14A9: to=<bahis@server.bahis.net>, relay=
local, delay=0.02, delays=0.01/0.01/0/0, dsn=2.0.0, status=sent (delivered to mailbox)
Feb 11 20:34:16 server postfix/qmgr[10155]: 794CE14A9: removed
^C
[root@server.bahis.net ~]# tail -f /var/log/maillog
Feb 11 20:33:43 server postfix/postfix-script[10151]: starting the Postfix mail system
Feb 11 20:33:43 server postfix/master[10153]: daemon started -- version 3.5.25, configurati
on /etc/postfix
Feb 11 20:34:16 server postfix/smtpd[10156]: warning: run-time library vs. compile-time hea
der version mismatch: OpenSSL 3.5.0 may not be compatible with OpenSSL 3.2.0
```

16:

2.14

-
- : Received: from client.bahis.net (192.168.1.30)
- : status=sent (delivered to mailbox)

```
Feb 11 20:34:16 server postfix/smtpd[10156]: disconnect from client.bahis.net[192.168.1.30]
 ehlo=2 starttls=1 mail=1 rcpt=1 data=1 quit=1 commands=7
Feb 11 20:34:16 server postfix/local[10161]: 794CE14A9: to=<bahis@server.bahis.net>, relay=
local, delay=0.02, delays=0.01/0.01/0/0, dsn=2.0.0, status=sent (delivered to mailbox)
Feb 11 20:34:16 server postfix/qmgr[10155]: 794CE14A9: removed
^C
[root@server.bahis.net ~]# tail -f /var/log/maillog
Feb 11 20:33:43 server postfix/postfix-script[10151]: starting the Postfix mail system
Feb 11 20:33:43 server postfix/master[10153]: daemon started -- version 3.5.25, configurati
on /etc/postfix
Feb 11 20:34:16 server postfix/smtpd[10156]: warning: run-time library vs. compile-time hea
der version mismatch: OpenSSL 3.5.0 may not be compatible with OpenSSL 3.2.0
```

16:

2.15 DNS- MX-

- MX 10 mail.bahis.net.

```
$TTL 86400
@   IN  SOA server.bahis.net. root.bahis.net. (
        2026020109
        86400
        3600
        604800
        10800 )

    IN  NS  server.bahis.net.
    IN  A   192.168.1.1
    IN  MX 10 mail.bahis.net.
server  IN  A   192.168.1.1
ns      IN  A   192.168.1.1
dhcp    IN  A   192.168.1.1
www     IN  A   192.168.1.1
```

2.15 DNS- MX-

- MX 10 mail.bahis.net.
- A- mail → 192.168.1.1

```
$TTL 86400
@   IN  SOA server.bahis.net. root.bahis.net. (
        2026020109
        86400
        3600
        604800
        10800 )

    IN  NS  server.bahis.net.
    IN  A   192.168.1.1
    IN  MX 10 mail.bahis.net.
server  IN  A   192.168.1.1
ns      IN  A   192.168.1.1
dhcp    IN  A   192.168.1.1
www     IN  A   192.168.1.1
```

2.15 DNS- MX-

- MX 10 mail.bahis.net.
- A- mail → 192.168.1.1
-

```
$TTL 86400
@   IN  SOA server.bahis.net. root.bahis.net. (
        2026020109
        86400
        3600
        604800
        10800 )

    IN  NS  server.bahis.net.
    IN  A   192.168.1.1
    IN  MX 10 mail.bahis.net.
server  IN  A   192.168.1.1
ns      IN  A   192.168.1.1
dhcp    IN  A   192.168.1.1
www     IN  A   192.168.1.1
```

2.16 DNS-

- PTR- 192.168.1.1

```
$TTL 86400
@   IN  SOA server.bahis.net. root.bahis.net. (
        202602010
        86400
        3600
        604800
        10800 )

    IN  NS  server.bahis.net.
    IN  PTR server.bahis.net.
    IN  MX 10 mail.bahis.net.
1   IN  PTR server.bahis.net.
1   IN  PTR ns.bahis.net.
1   IN  PTR dhcp.bahis.net.
1   IN  PTR www.bahis.net.
30  IN  PTR client.bahis.net.
1   IN  PTR mail.bahis.net.
```

18: DNS- PTR-

2.16 DNS-

- PTR- 192.168.1.1
- : server, ns, dhcp, www, mail

```
$TTL 86400
@   IN  SOA server.bahis.net. root.bahis.net. (
        202602010
        86400
        3600
        604800
        10800 )

    IN  NS  server.bahis.net.
    IN  PTR server.bahis.net.
    IN  MX 10 mail.bahis.net.
1   IN  PTR server.bahis.net.
1   IN  PTR ns.bahis.net.
1   IN  PTR dhcp.bahis.net.
1   IN  PTR www.bahis.net.
30  IN  PTR client.bahis.net.
1   IN  PTR mail.bahis.net.
```

18: DNS- PTR-

2.16 DNS-

- PTR- 192.168.1.1
- : server, ns, dhcp, www, mail
-

```
$TTL 86400
@   IN  SOA server.bahis.net. root.bahis.net. (
        202602010
        86400
        3600
        604800
        10800 )

    IN  NS  server.bahis.net.
    IN  PTR server.bahis.net.
    IN  MX 10 mail.bahis.net.
1   IN  PTR server.bahis.net.
1   IN  PTR ns.bahis.net.
1   IN  PTR dhcp.bahis.net.
1   IN  PTR www.bahis.net.
30  IN  PTR client.bahis.net.
1   IN  PTR mail.bahis.net.
```

18: DNS- PTR-

mydestination

- mydestination

```
[root@server.bahis.net ~]# postconf -e 'inet_interfaces = all'
[root@server.bahis.net ~]# postconf -e 'mynetworks = 127.0.0.0/8, 192.168.0.0/16'
[root@server.bahis.net ~]# postfix check
[root@server.bahis.net ~]# systemctl reload postfix
[root@server.bahis.net ~]# systemctl stop postfix
[root@server.bahis.net ~]# systemctl start postfix
[root@server.bahis.net ~]# cat /var/spool/mail/bahis
```

19: mydestination

mydestination

- mydestination
- postfix check systemctl reload postfix

```
[root@server.bahis.net ~]# postconf -e 'inet_interfaces = all'
[root@server.bahis.net ~]# postconf -e 'mynetworks = 127.0.0.0/8, 192.168.0.0/16'
[root@server.bahis.net ~]# postfix check
[root@server.bahis.net ~]# systemctl reload postfix
[root@server.bahis.net ~]# systemctl stop postfix
[root@server.bahis.net ~]# systemctl start postfix
[root@server.bahis.net ~]# cat /var/spool/mail/bahis
```

19: mydestination

mydestination

- mydestination
- postfix check systemctl reload postfix
- SELinux

```
[root@server.bahis.net ~]# postconf -e 'inet_interfaces = all'
[root@server.bahis.net ~]# postconf -e 'mynetworks = 127.0.0.0/8, 192.168.0.0/16'
[root@server.bahis.net ~]# postfix check
[root@server.bahis.net ~]# systemctl reload postfix
[root@server.bahis.net ~]# systemctl stop postfix
[root@server.bahis.net ~]# systemctl start postfix
[root@server.bahis.net ~]# cat /var/spool/mail/bahis
```

19: mydestination

mydestination

- mydestination
- postfix check systemctl reload postfix
- SELinux
- named

```
[root@server.bahis.net ~]# postconf -e 'inet_interfaces = all'
[root@server.bahis.net ~]# postconf -e 'mynetworks = 127.0.0.0/8, 192.168.0.0/16'
[root@server.bahis.net ~]# postfix check
[root@server.bahis.net ~]# systemctl reload postfix
[root@server.bahis.net ~]# systemctl stop postfix
[root@server.bahis.net ~]# systemctl start postfix
[root@server.bahis.net ~]# cat /var/spool/mail/bahis
```

19: mydestination

mydestination

- mydestination
- postfix check systemctl reload postfix
- SELinux
- named
- postqueue -f

```
[root@server.bahis.net ~]# postconf -e 'inet_interfaces = all'
[root@server.bahis.net ~]# postconf -e 'mynetworks = 127.0.0.0/8, 192.168.0.0/16'
[root@server.bahis.net ~]# postfix check
[root@server.bahis.net ~]# systemctl reload postfix
[root@server.bahis.net ~]# systemctl stop postfix
[root@server.bahis.net ~]# systemctl start postfix
[root@server.bahis.net ~]# cat /var/spool/mail/bahis
```

19: mydestination

2.18

- `test3  bahis@bahis.net`

2.19

- `status=sent (delivered to mailbox)`

```
        id 7C05114F3; Wed, 11 Feb 2026 19:49:35 +0000 (UTC)
Date: Wed, 11 Feb 2026 19:49:35 +0000
To: bahis@server.bahis.net
Subject: test1
User-Agent: s-nail v14.9.22
Message-Id: <20260211194935.7C05114F3@server.bahis.net>
From: root <root@server.bahis.net>
```

20: Postfix

2.19

- status=sent (delivered to mailbox)
- MX mydestination

```
        id 7C05114F3; Wed, 11 Feb 2026 19:49:35 +0000 (UTC)
Date: Wed, 11 Feb 2026 19:49:35 +0000
To: bahis@server.bahis.net
Subject: test1
User-Agent: s-nail v14.9.22
Message-Id: <20260211194935.7C05114F3@server.bahis.net>
From: root <root@server.bahis.net>
```

20: Postfix

2.20 DNS provision

- provision

```
[root@server.bahis.net ~]# cd /vagrant/provision/server/dns/var/named
[root@server.bahis.net named]# cp -R /var/named/* /vagrant/provision/server/dns/var/named
cp: overwrite '/vagrant/provision/server/dns/var/named/data/named.run'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/dynamic/managed-keys.bind.jnl'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/dynamic/managed-keys.bind'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/master/fz/bahis.net'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/master/rz/192.168.1'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/named.ca'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/named.empty'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/named.localhost'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/named.loopback'? y
[root@server.bahis.net named]# cd /vagrant/provision/server
[root@server.bahis.net server]# touch mail.sh
```

21: DNS provision

2.20 DNS provision

- provision
- /var/named

```
[root@server.bahis.net ~]# cd /vagrant/provision/server/dns/var/named
[root@server.bahis.net named]# cp -R /var/named/* /vagrant/provision/server/dns/var/named
cp: overwrite '/vagrant/provision/server/dns/var/named/data/named.run'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/dynamic/managed-keys.bind.jnl'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/dynamic/managed-keys.bind'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/master/fz/bahis.net'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/master/rz/192.168.1'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/named.ca'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/named.empty'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/named.localhost'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/named.loopback'? y
[root@server.bahis.net named]# cd /vagrant/provision/server
[root@server.bahis.net server]# touch mail.sh
```

21: DNS provision

2.20 DNS provision

- provision
- /var/named
-

```
[root@server.bahis.net ~]# cd /vagrant/provision/server/dns/var/named
[root@server.bahis.net named]# cp -R /var/named/* /vagrant/provision/server/dns/var/named
cp: overwrite '/vagrant/provision/server/dns/var/named/data/named.run'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/dynamic/managed-keys.bind.jnl'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/dynamic/managed-keys.bind'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/master/fz/bahis.net'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/master/rz/192.168.1'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/named.ca'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/named.empty'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/named.localhost'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/named.loopback'? y
[root@server.bahis.net named]# cd /vagrant/provision/server
[root@server.bahis.net server]# touch mail.sh
```

21: DNS provision

2.21 mail.sh

- `mail.sh`

2.21 mail.sh

- mail.sh
- postfix s-nail

2.21 mail.sh

- `mail.sh`
- `postfix` `s-nail`
- firewall SELinux

2.21 mail.sh

- mail.sh
- postfix s-nail
- firewall SELinux
- mydomain, myorigin, inet_protocols, inet_interfaces, mynetworks, mydestination

2.21 mail.sh

- mail.sh
- postfix s-nail
- firewall SELinux
- mydomain, myorigin, inet_protocols, inet_interfaces, mynetworks, mydestination
-

2.22 mail.sh

- `mail.sh` provision

```
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y install postfix
dnf -y install s-nail
echo "Configure postfix"
postconf -e 'inet_protocols = ipv4'
echo "Start postfix service"
```

2.22 mail.sh

- `mail.sh` provision
-

```
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y install postfix
dnf -y install s-nail
echo "Configure postfix"
postconf -e 'inet_protocols = ipv4'
echo "Start postfix service"
```

2.22 mail.sh

- `mail.sh` provision
-
- `postfix` `s-nail`

```
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y install postfix
dnf -y install s-nail
echo "Configure postfix"
postconf -e 'inet_protocols = ipv4'
echo "Start postfix service"
```

2.22 mail.sh

- mail.sh provision
-
- postfix s-nail
- inet_protocols = ipv4

```
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y install postfix
dnf -y install s-nail
echo "Configure postfix"
postconf -e 'inet_protocols = ipv4'
echo "Start postfix service"
```

2.22 mail.sh

- `mail.sh` provision
-
- `postfix` `s-nail`
- `inet_protocols = ipv4`
-

```
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y install postfix
dnf -y install s-nail
echo "Configure postfix"
postconf -e 'inet_protocols = ipv4'
echo "Start postfix service"
```

2.22 mail.sh

- `mail.sh` provision
-
- `postfix s-nail`
- `inet_protocols = ipv4`
-
- Postfix

```
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y install postfix
dnf -y install s-nail
echo "Configure postfix"
postconf -e 'inet_protocols = ipv4'
echo "Start postfix service"
```

2.23 provisioning- Vagrantfile

- `server.vm.provision`

```
server.vm.provision "server mail",
type: "shell",
preserve_order: true,
path: "provision/server/mail.sh"
```

23: provisioning Vagrantfile

2.23 provisioning- Vagrantfile

- server.vm.provision
- provision/server/mail.sh

```
server.vm.provision "server mail",
type: "shell",
preserve_order: true,
path: "provision/server/mail.sh"
```

23: provisioning Vagrantfile

2.23 provisioning- Vagrantfile

- server.vm.provision
- provision/server/mail.sh
-

```
server.vm.provision "server mail",
type: "shell",
preserve_order: true,
path: "provision/server/mail.sh"
```

23: provisioning Vagrantfile

2.24 provisioning- Vagrantfile

- `client.vm.provision`

```
client.vm.provision "client mail",
type: "shell",
preserve_order: true,
path: "provision/client/mail.sh"
```

24: provisioning Vagrantfile

2.24 provisioning- Vagrantfile

- client.vm.provision
- provision/client/mail.sh

```
client.vm.provision "client mail",
type: "shell",
preserve_order: true,
path: "provision/client/mail.sh"
```

24: provisioning Vagrantfile

2.24 provisioning- Vagrantfile

- client.vm.provision
- provision/client/mail.sh
-

```
client.vm.provision "client mail",
type: "shell",
preserve_order: true,
path: "provision/client/mail.sh"
```

24: provisioning Vagrantfile

3

3.

3.1

- Postfix `bahis.net`

3.1

- Postfix `bahis.net`
-

3.1

- Postfix `bahis.net`
-
-

3.1

- Postfix `bahis.net`
-
-
- MX PTR- DNS

3.1

- Postfix `bahis.net`
-
-
- MX PTR- DNS
-

3.1

- Postfix `bahis.net`
-
-
- MX PTR- DNS
-
- provisioning-